Substitution

Example

If a = 4, b = 2 and c = 3, what are the values of the following expressions?

a. $a + b$
b. $b - c$
c. $ab + c$

When we substitute, we replace something or someone with another thing or person. For example, during a football match, the coach may substitute a player by swapping them out with another already on the field.

For our substitution here, we need to substitute a **letter** for a **number**.

Step One

When we substitute, we replace the letter for its value.

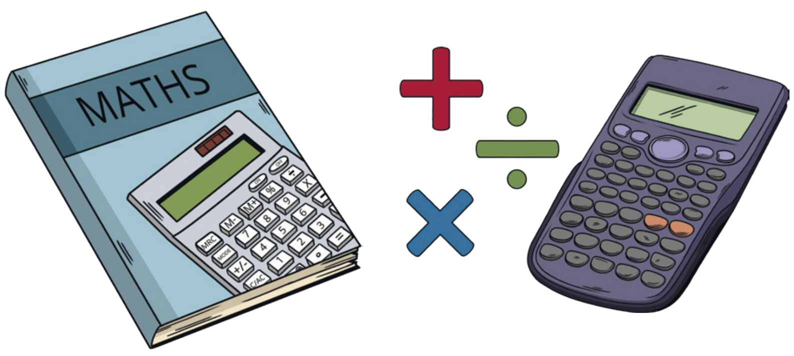
We are told that, in this example, a has a value of 4. Therefore, we can substitute a for 4.

a. $4 + b$
b. $b - c$
c. $4b + c$

Step Three

Substitute the letter c for its value.

a. 4 + 2
b. 2 - 3
c. 4 × 2 + 3 or 8 + 3



Step Two

Now substitute the letter b for its value. b has a value of 2:

a. 4 + 2
b. $2 - c$
c. $4b$ is slightly different; we cannot simply place the 2 next to the 4,

i.e. the answer is not $42 + c$.

Rather, we must remember there is an invisible multiplication sign between the 4 and b.

So, our substitution becomes 4 × 2.

$4 \times 2 + c$

which is equal to

$8 + c$

Step Four

You have substituted the values for the letters you have been given.

a. 4 + 2 = 6
b. 2 - 3 = -1
c. 8 + 3 = 11

BEYOND MATHS